ICT PROJECT / STAGE 1

Campus Lost & Found

A university service for lost and found items

UNIVERSITY ACCESS

Accounts at astanait.edu.kz

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SE-2623

Midterm defense
9 October 2026

CONCEPT & PROPOSED DESIGN

Problem, users and scope

Goal: help students find notices and contact the person who posted them.

Working hypothesis

Lost-item messages may be hard to find when students use separate channels.

User group	Need	First-version scope
Students	Find or return an item	Post, search, contact and close notices
Administrator	Keep notices relevant	Review notices and remove misuse

Outside the first version

Payments, live chat, delivery and automatic item matching.

Student survey | 10 responses

9/10 rated the service useful; 4/10 reported returns taking over 3 days. Convenience sample; not representative of all university students.

Sources: Student survey | Response summary (team access)

Functional requirements

Eight proposed functions. Each has a priority and an acceptance check.

ID	Function / priority	Acceptance check
F1	University account sign-in **Must**	Reject identities outside astanait.edu.kz
F2	Publish a lost or found notice **Must**	A valid notice appears in the list
F3	Attach one photo **Should**	Accept JPG/PNG files up to 5 MB
F4	Browse notices and open details **Must**	List and details show saved content
F5	Search title; filter by category **Must**	Seeded matching notices appear
F6	Contact via university email **Must**	Contact opens the author's email
F7	Edit own notice; mark returned **Must**	Owner changes persist after refresh
F8	Administrator removes misuse **Must**	Admin succeeds; student is denied

Top three priorities

University access, publishing and search form the core workflow.

Quality and success tests

All targets below are proposed acceptance criteria, not measured results.

ID / type	Target and check - all Must
N1 Performance	Search p95 <= 2 seconds with 10 concurrent users. **Check:** repeated searches on the demo laptop.
N2 Security	Enforce university identity and owner/admin rights. **Check:** invalid identities and forbidden actions.
N3 Usability	Support 360 px screens and keyboard input. **Check:** core screens, labels and focus order.
N4 Availability	Pass 20/20 checks during a 30-minute demo. **Check:** core pages plus retry when offline.

Three measurable success criteria

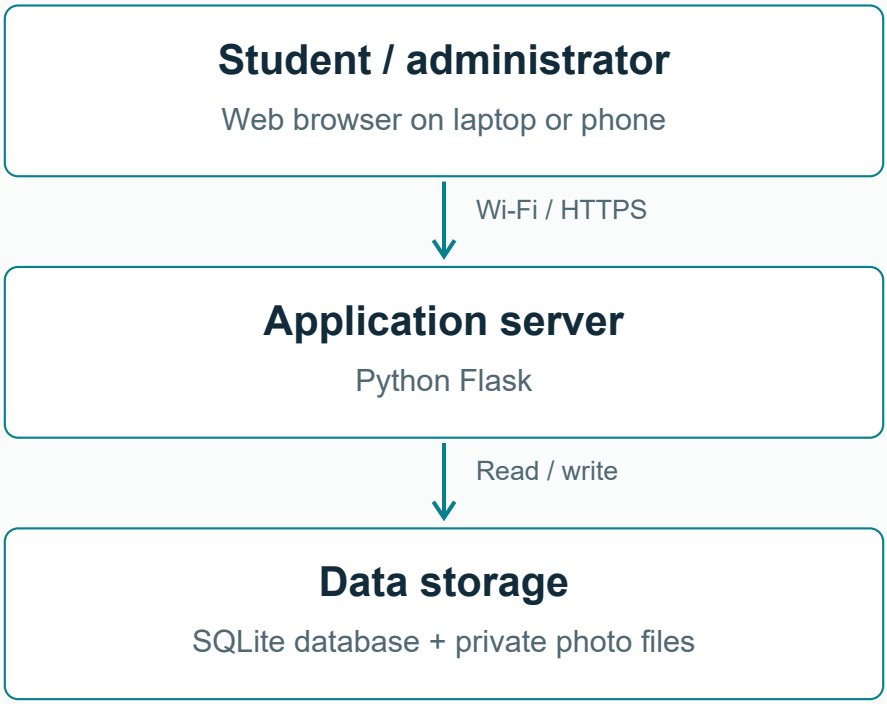
4/5 testers post a notice within 2 minutes

4/5 testers find an item within 1 minute

5/5 invalid-access checks are blocked

Architecture and technology

A small client-server prototype; deployment and university SSO need setup.



University identity provider
SSO configuration and university permission are pending.

Layer	Tools and reason
Interface	HTML, CSS, JavaScript - simple browser UI
Server / data	Flask, SQLite, private files - small local prototype
Team workflow	VS Code, Git, Google Drive - history and shared files

Planned deployed data flow

DNS resolves the host. HTTPS carries the request. Flask checks access, reads storage and returns the result.

Stored data: planned disk encryption and restricted file access.

Sources: Flask tutorial: database

Cloud service choice

Google Drive supports team documents. Application data stays on the prototype server.

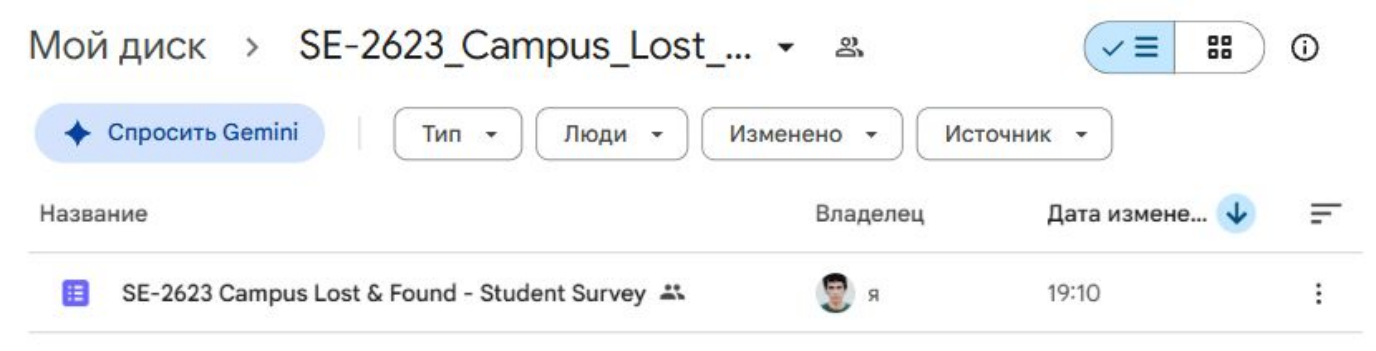
Service	Free personal plan	Collaboration / platform fit
Google Drive	Up to 15 GB Shared with Gmail and Photos	Docs / Slides co-editing. Browser, desktop, mobile.
OneDrive	5 GB Microsoft cloud storage	Word / PowerPoint collaboration. Browser, Windows, Mac, mobile.
iCloud Drive	5 GB Shared iCloud storage	Pages / Keynote collaboration. Apple tools and browser access.

Selected: Google Drive

Shared browser editing and its free allowance fit our project materials.

All three support two-factor authentication and encryption in transit and at rest. Paid pricing varies by region; university quotas may differ.

Real project folder - open in Google Drive



Owner + two editors; access restricted to the team. Student survey saved inside.
Real folder screenshot, captured 6 October 2026.

Sources: Google quota | Drive sharing | OneDrive plans | iCloud plans | Google security
OneDrive security | Apple security

Student screen

Task: find a relevant notice and contact its author.

Campus Lost & Found

Signed in: university account

Search items...

Category: All

Item	Status	Location
Blue water bottle	Found	Library
Black headphones	Lost	Block C

Open notice: photo, date, description and contact

Post a notice

Illustrative items. This is a wireframe.

01 Visibility of system status

Lost / Found / Returned labels make the notice state visible.

02 Recognition rather than recall

Visible categories and item details support quick scanning.

Accessibility target

44 px controls, clear labels, visible keyboard focus and strong contrast.

Sources: Nielsen: 10 usability heuristics

Administrator screen

Task: review notices and remove inappropriate content.

Moderation

Signed in: administrator

Notice	Observed issue	Action
Headphones notice	Duplicate	Review
Item sale notice	Outside scope	Review

Selected notice: Item sale notice

Remove this notice?

This notice will be removed from the list.

Cancel | **Confirm removal**

Illustrative notices and administrator-observed issues.

01 Error prevention

Show the selected notice and ask for confirmation before removal.

02 User control and freedom

Cancel returns to the list without removing anything.

Accessibility target

Keyboard navigation and a visible focus indicator on every action.

Sources: Nielsen: 10 usability heuristics

Stage 2 work plan

All three members build and review together. A weekly lead coordinates the work.

Week / lead	Planned work	Deliverable
6 **Zhanzhol + team**	Finalize the data model and forms. Request SSO configuration.	Local skeleton and identity decision
7 **Daniil + team**	Build posting, search and university-email contact.	Core notice workflow
8 **Daulet + team**	Add moderation and photos. Run usability and access checks.	Tested student and admin flows
9 **All three**	Fix issues, rehearse and prepare the demonstration.	Working local prototype and test summary

Week 9 demonstration

Create a notice, search for it, contact its author, mark it returned and moderate misuse.

Risks and shared preparation

One shared project: every team member reviews the design and answers questions.

Risk	Likelihood / impact	Response
SSO approval is delayed	Medium / High	Ask IT in Week 6. Clearly label simulated sign-in if needed for the demo.
Scope grows beyond available time	High / Medium	Keep posting and search first. Defer photo upload if needed.
Files are unavailable during the defense	Medium / High	Keep evidence in the shared folder and download an offline copy before 9 October.

Team contributions

Team member	Contribution
Mengdigerey Zhanzhol	Part A: problem and users Part B: functional requirements
Galitskiy Daniil	Part B: non-functional requirements Part C: architecture and cloud
Bisengaliev Daulet	Part D: interface design Part E: plan and risks

All three members reviewed the project and prepared the presentation together.